Running this provider under ADCS: a readiness plan

Assessment of this provider against what an Active Directory Certificate Services certification authority requires, and the phased plan to close the difference. Generated 2026-10-18 from the markdown of record.

The question. Can this KSP back an Active Directory Certificate Services certification authority — the CA's private key living on a PKCS#11 token, reached through this provider?

The verdict. The architecture is right for it and nothing found so far says otherwise. Today it cannot work, for three reasons that have nothing to do with ADCS, plus four things genuinely missing for the CA case and two that are design decisions rather than code.

This document separates those, because they have completely different costs: one needs a machine, two need a decision from whoever owns the CA's threat model, and four are ordinary engineering.

What a CA actually asks of a KSP

Checked against the source rather than assumed. The point of this table is that the functional surface is largely already there — the problems are elsewhere.

What certsrv needs	State	Evidence
Register under the key-storage interface	**Present**	**tools/register_ksp.ps1** uses **NCRYPT_KEY_STORAGE_INTERFACE** with a user-mode image
Machine-scoped key generation	**Present**	**NCRYPT_MACHINE_KEY_FLAG** honoured in **KSP_CreatePersistedKey**
RSA signing, PKCS#1 v1.5 and PSS	**Present**	verified against two real tokens
ECDSA signing, P-256/384/521	**Present**	verified against two real tokens
Open the CA key by container name	**Present**	**KSP_OpenKey**, 256-byte label budget — ample for a CA name
Enumerate keys	**Present**	**KSP_EnumKeys**, scope-filtered
Store the CA certificate on the key	**Present**	**NCRYPT_CERTIFICATE_PROPERTY** (PROP-14)
Non-exportable private key	**Present, and the default**	**CKA_SENSITIVE** + **CKA_EXTRACTABLE=FALSE**
Advertise algorithms and key sizes	**Half present**	**EnumAlgorithms** yes; **NCRYPT_LENGTHS_PROPERTY missing** — see G3
Tolerate **NCRYPT_SILENT_FLAG**	**Present**	flags are masked with **&**, never compared whole, so **SILENT|PAD_PKCS1** resolves correctly (**p11_utils.c:248**). Benign by construction, but untested — see A4
Survive a multithreaded service	**Present**	32 threads over the 16-session pool, under ThreadSanitizer
Set a DACL on the CA key	**Absent**	**NCRYPT_SECURITY_DESCR_PROPERTY** not implemented — see G6
Refuse a duplicate container name	**Absent**	no **NTE_EXISTS**, no **NCRYPT_OVERWRITE_KEY_FLAG** — see G4
Authenticate without a prompt	**Present but wrong for a CA**	PIN from an environment variable — see G5

The gaps, in the order they bite

G1 — There is no DLL. *Blocking everything.*

ksp_main.c cannot compile without **ncrypt_provider.h**, which declares

NCRYPT_KEY_STORAGE_FUNCTION_TABLE. MSVC compiles the other nine source files clean at **/W3 /WX**; CMake skips the DLL target rather than emitting a provider with no entry point.

Not an ADCS problem. It is *the* problem, and it gates every other phase.

G2 — The binary is unsigned. *Probably blocking.*

tools/sign_ksp.ps1 signs, timestamps and verifies, and CI signs automatically once the certificate secret exists. What is missing is an OV or EV code-signing certificate, which a verified legal entity has to buy.

> **Unverified claim, and it matters:** I expect **ncrypt.dll** to refuse to load an unsigned user-mode KSP. I have not confirmed that and this repository does not either. Settle it in the first hour on Windows (step B1) — if signing is required, it is a procurement lead time, not a code change, and it should start immediately.

G3 — NCRYPT_LENGTHS_PROPERTY was not implemented. *Closed in Phase A.*

Every shipping KSP implements it. It returns **NCRYPT_SUPPORTED_LENGTHS** (minimum, maximum, increment, default), and it is what the CA setup wizard and **certutil** read to populate and validate key-size choices. The provider answers **NCRYPT_LENGTH_PROPERTY** — the singular, a given key's own size — and has nothing for the plural.

> **The matrix never noticed**, and the reason is instructive: **PROP-02** tracks the singular and there is no row at all for the plural. Two properties one letter apart, one audited and one invisible. This is the same blind spot that hid the two stubs and the missing **NTE_EXISTS**, and the third instance in three sessions.

Cheap to close: the provider already holds **KSP_RSA_MIN_BITS**,

KSP_RSA_MAX_BITS and the step-64 rule, and the EC sizes are per curve.

G4 — A reused container name silently duplicates the key. *Closed.*

> **Done in Phase A.** **NCryptCreatePersistedKey** now returns **NTE_EXISTS** for a name already in use, and **NCRYPT_OVERWRITE_KEY_FLAG** replaces deliberately. The replaced objects — private, public, secret and the certificate stored beside them — are destroyed **only after the new key has generated**, so a failed generation leaves the key it was replacing intact. Destroying first and generating second would lose a CA key outright when generation fails, which is unrecoverable; generating first costs a moment in which two objects share a label, and that is the better trade. Both directions and the ordering are fault-injected.

> **Still to confirm on Windows (E4):** which call pattern ADCS actually uses on renewal — whether it passes the overwrite flag, derives a fresh name, or expects **NTE_EXISTS** and handles it. That decides whether the refusal or the overwrite is the path that runs, not whether either is correct.

The original description follows.

NCryptCreatePersistedKey does not check whether the name is taken and does not handle **NCRYPT_OVERWRITE_KEY_FLAG**, so two token objects can carry the same **CKA_LABEL** and **KSP_OpenKey** returns whichever the token's search hands back first.

For most callers that is untidy. **For a CA it is a correctness failure with certificate-validity consequences**: renewing a CA certificate is exactly the operation that reuses or re-derives a container name, and a CA that signs with an indeterminate key generation issues certificates that may not chain.

Found while making the two-token suite repeatable, two sessions ago, and deferred then as "its own change". The ADCS target promotes it.

Still to confirm: the exact call pattern ADCS uses on renewal — whether it passes
NCRYPT_OVERWRITE_KEY_FLAG, derives a new name, or expects

NTE_EXISTS. That needs a Windows box (step D4) and it decides whether the fix is a refusal, an overwrite, or both.

G5 — PIN delivery. *Partly closed; the delivery mechanism is still a decision.*

> **The silent default is gone** (A5): no PIN configured is now **NTE_PERM** and a log line, not an attempt with **"1234"**. What remains is the question below — how a headless **certsrv** is given a real credential — and that still needs an owner's decision rather than a patch.

certsrv is a headless service. It never calls

NCryptSetProperty(NCRYPT_PIN_PROPERTY), so the PIN comes from

GetEffectivePin, which reads an environment variable. For a service that means a **machine-wide** variable, readable by any process on the host.

*(This paragraph used to end "and **config.h:36** defines a fallback of

1234 when nothing is set" — contradicting the paragraph above it in the same section, which records A5 removing exactly that. Corrected in session 19.)*

A machine-wide variable inverts the security model a CA is bought for. The key cannot be extracted from the token — good — but anyone on that host can ask the token to sign, and for a CA, signing *is* the asset.

> **Two of the four options are now implemented, and they are the two that remove the problem rather than relocating it.** The provider used to demand a PIN of every token; it now asks the token how it wants to be authenticated and honours all three PKCS#11 models — see Phase C. That leaves a decision only if your token turns out to need a PIN.

G6 — No ACL on the key. *A design decision.*

NCRYPT_SECURITY_DESCR_PROPERTY is not implemented because PKCS#11 has no ACL model (matrix **PROP-11**, Not covered). The Microsoft Software KSP protects machine keys with a filesystem DACL, and the CA MMC's private-key permissions tooling expects to read and set one.

Consequence: you cannot restrict the CA key to the CA's service account through Windows. The only boundary available is the token's own, which Phase D turns into the mitigation.

G7 — The Windows and conformance suites have never executed.

40 integration tests and roughly 150 HLK-conformant checks, written and syntax-checked on Linux only. A suite that has never run is a hypothesis about the suite as much as about the code.

G8 — SoftHSM2 is a software token.

Encrypted SQLite, no hardware protection. Correct for a lab CA; not what you want under a production root or issuing CA. The provider is backend-agnostic, so the production path is a real HSM's PKCS#11 library.

Note the economics honestly: an HSM that ships its own KSP does not need this one. The value here is an HSM *without* a Windows KSP, a token like Kryoptic, or a test environment.

The plan

Five phases. **Phase A is the only one that can be done from this environment**, which is why it comes first — it stops the scarce Windows time being spent on things that did not need Windows.

Phase A — Close what can be closed without Windows

	Work	Why now
A1	**Done.** `NCRYPT_LENGTHS_PROPERTY` as `NCRYPT_SUPPORTED_LENGTHS`: a range for RSA and AES, one fixed size with **`dwIncrement = 0`** for curves, EdDSA, HMAC and ML-DSA. Matrix row **`PROP-19`** added.	G3. The CA wizard reads it; cheap; no Windows needed to get it right
A2	**Done. `NTE_EXISTS`** on a taken container name, plus **`NCRYPT_OVERWRITE_KEY_FLAG`** to replace deliberately, destroying the replaced objects only after the new key generates. Both directions and the ordering fault-injected.	G4. Had to land before any renewal test, or that test could not be trusted
A3	**Done**, and it found a defect on its first run — see below. **`tests/linux/test_adcs_lifecycle.c`**, 42 assertions, run twice over one token, plus a no-PIN case in its own process.	De-risked the whole CA flow against a real token before Windows exists
A4	A regression test that **`NCRYPT_SILENT_FLAG`** combined with each padding flag still resolves the right mechanism.	The masking is correct today *by construction*; nothing stops a future **`==`** comparison breaking CA signing
A5	**Done**, and for BOTH scopes rather than machine only: a scope-dependent default would mean a developer's user-scope key silently works while the identical machine-scope configuration fails. **`NTE_PERM`** plus a log line naming all three places a PIN can come from; **`-DKSP_ALLOW_DEFAULT_PIN=1`** restores it for a lab.	G5, the part that needs no decision

Gate: A1–A5 green, fault-injected, and the live suite passing twice.

What A3 found on its first run

A key opened by name never recovered its RSA modulus size, so every signature with it failed at the first of CNG's two calls.

`KSP_OpenKey` read **`CKA_MODULUS_BITS`** from the **private** key. PKCS#11 v2.40 puts that attribute on the **public** key; the private key carries

`CKA_MODULUS` but not its bit count. The read's return value was ignored, so **`dwKeyBitLen`** silently became 0.

Nothing noticed because a zero length is invisible to the operation that looks most important. Signing passes a real buffer and the token knows its own key size, so signatures came out correct. It surfaces only on the **size query** — **`KspSignatureLength`** refuses a zero-bit key — and CNG calls every operation twice, length first. **So every signature with a key opened by name would have failed at the first call, which for a CA is every operation after setup.** SoftHSM2 tolerates the misdirected read, which is why five sessions of live testing against it never saw this.

Fixed by reading the attribute from the public key where the spec puts it, falling back to the length of **`CKA_MODULUS`** on the private key, and refusing the open outright if neither works rather than returning a key whose size is unknown. Re-injecting the original read reproduces the failure.

This is the fifth time a defect has survived the mock and been found by a real token, and the second time the cause was an attribute read from the wrong object class. It is also the argument for Phase B being a gate rather than a formality: the same class of mistake is waiting in everything

`ncrypt.dll` does that no suite here has ever exercised.

Phase B — Prove the artifact exists *(needs one Windows machine)*

	Work
B1	Obtain **ncrypt_provider.h** (Cryptographic Provider Development Kit — association unconfirmed; two guesses at its location were already wrong). Build the DLL. **Also settle G2's open question immediately: does an unsigned KSP load at all?**
B2	Register with **register_ksp.ps1**; confirm **certutil -csplist** lists it
B3	Run the 40 integration tests and the ~150 conformance checks. Expect failures; this is their first execution
B4	Confirm the function table layout against the real header, closing **TABLE-01**

Gate: the provider loads, enumerates, and signs through the real

ncrypt.dll. Until this gate passes, every ADCS statement is a projection.

Phase C — How the CA authenticates to the token

C-c and C-d are implemented (session 19). The provider no longer assumes a PIN: it reads **CK_TOKEN_INFO.flags** for the scope's slot and honours all three models PKCS#11 describes.

Option	How	Status
C-a Machine-wide env var	Status quo	Works. Any process on the host reads the PIN — fine for a lab, **not for production**
C-b DPAPI machine-scope blob	PIN encrypted with **CryptProtectData** at machine scope, in a registry value ACL'd to the CA service account; provider decrypts at initialisation	**Not implemented.** Windows-only, so the Linux suites cannot cover it — it needs its own Windows test, which means it needs Phase B first
C-c Protected authentication path	Token sets **CKF_PROTECTED_AUTHENTICATION_PATH**; provider calls **C_Login** with **pPin = NULL** per v2.40 §11.6, and **requires no configured PIN**	**Implemented and tested.** Both directions fault-injected
C-d Token wants no login	Token leaves **CKF_LOGIN_REQUIRED** clear; provider performs no **C_Login** at all	**Implemented and tested**

This was not only a missing feature — it was a defect. Before this, the provider demanded a PIN of every token *and* refused to start without one (A5's fail-closed rule). A hardware HSM with a PIN pad — exactly the certified-HSM case this project targets — was therefore **rejected outright**, with **NTE_PERM**, for not having a PIN configured that it did not need. **CKF_PROTECTED_AUTHENTICATION_PATH** and **CKF_LOGIN_REQUIRED** were not even defined in this project's **pkcs11.h**.

The fail-closed rule now applies **only** where it belongs: a token that actually asked for a PIN. A **C_GetTokenInfo** that fails falls through to the strict PIN model rather than guessing something laxer — an unreadable answer must not be the reason a credential stops being required, and that direction is fault-injected too.

What is left to decide: only if your token requires a PIN. Run

tools/p11_qualify against it — it reports which of the three models the token uses, so this is answered by measurement rather than by a datasheet. If it does need a PIN, C-b is the recommendation and it needs Phase B first. If you are buying an HSM, **ask the vendor for C-c**: it is now supported here, and it removes the credential from Windows entirely.

Phase D — Protect the key without a Windows ACL

The honest position is that this provider cannot offer a DACL, so the boundary has to be the token's.

	Work
D1	**Use per-scope tokens, which already exist.** `KSP_MACHINE_TOKEN_LABEL` + `KSP_MACHINE_PIN` give the CA its own token with its own credential and its own session pool — built in proposal C and verified against two real tokens with different PINs. This is the strongest lever available and it is already implemented
D2	Document plainly that **NCRYPT_SECURITY_DESCR_PROPERTY** is unavailable and that the CA MMC's private-key permissions dialog will not work. A CA operator must not discover that during an incident
D3	Decide whether an unavailable DACL is acceptable for this deployment, or whether it rules out the approach. **This is a risk acceptance, not an engineering task**

Phase E — The ADCS pilot *(needs Windows, after B and A)*

The test plan, in the order that finds problems soonest:

	Test	Watching for
E1	**certutil -csplist** shows the provider; the CA setup wizard offers it and populates key sizes	G3 — the wizard's key-length list is the **NCRYPT_LENGTHS_PROPERTY** test
E2	Install a **standalone test CA** with **-CryptoProviderName "SoftHSM KSP"**, RSA-4096, SHA-256	Key creation under the service account; PIN reaching a headless process
E3	Issue a certificate; publish a CRL	The main signing path through **ncrypt.dll**
E4	**Renew the CA certificate, with a new key**	G4. The test most likely to fail, and the one A2 exists to make meaningful
E5	Restart the CA service; then reboot the host	Lazy initialisation, PIN availability at boot before any logon
E6	Revoke; verify chains with **certutil -verify**	End-to-end correctness, not just API success
E7	Two CAs on one host, or a CA alongside another KSP consumer	The session pool and per-process singleton under a second consumer

Throw the test CA away afterwards. Never pilot against a CA whose certificates anything trusts.

Phase F — Production

Only meaningful once E passes: the code-signing certificate (G2), a real HSM rather than SoftHSM2 (G8), FIPS posture if required (**OPS-02**, needs certified hardware), and HA, which vendor PKCS#11 libraries handle themselves.

Effort, with a warning about these numbers

> **This project's effort estimates were wrong on 4 of the 4 rows it has ever closed.** Two were marked N/A — "no remedy planned" — and were then closed in a day. So read the sizes below as relative, not as a schedule.

Phase	Size	Can it start now?
A	days	**Yes, entirely**
B	hours of work, unknown lead time to obtain the header	No — needs a machine and a header
C	days, plus a decision	Decision now; code after B
D	D1 is done; D2 a page; D3 a decision	Yes
E	days of iteration	No — needs A and B
F	procurement and hardware	Partly now (start the certificate purchase)

The critical path is **the header**, and it has no engineering content. Start looking for it in parallel with Phase A.

What needs a decision from you

1. **Phase C option** — now only a decision *if your token requires a PIN*. C-c (protected authentication path) and C-d (no login) are implemented and tested; run **tools/p11_qualify** against the token to see which model it uses. If it does need a PIN, I recommend **C-b (DPAPI)**, which needs Phase B first because it is Windows-only code.
2. **Phase D3** — is a CA key with no Windows DACL, protected only by the token boundary and its PIN, acceptable for this deployment?
3. **Target token** — SoftHSM2 (lab only) or a specific HSM? It changes C and G8, and if the HSM ships its own KSP it may change whether this project is the right tool at all.
4. **Signing certificate** — start the purchase now if G2 turns out to be a hard requirement, because it has the longest lead time of anything here.

What this plan does not claim

- That the DLL builds. Nobody has built it.
- That ADCS accepts this provider. No version of it has ever been loaded by **ncrypt.dll**.
- That the ADCS call sequence is fully known. G4's resolution depends on observing it (E4), and some of what is written above about the wizard and renewal is inference from the CNG contract rather than observation.
- That Phase A's work is sufficient. It is what can be established without Windows, which is a different and smaller claim.

Everything above rests on reading this repository's source. Nothing in it has been tested against ADCS, because no artifact exists to test.

`Content fingerprint: 092b18f3392e` — of the markdown this was rendered from, plus this script. `tests/check_generated_artifacts.py` recomputes it, so a plan edited without regenerating is a CI failure rather than a PDF that quietly describes an older plan.